

AI Job Copilot V2.0 Release Report

Objective

Improve reliability, performance, error handling, logging, health checks, admin monitoring, rate limiting, idempotency, and document generation stability without changing the user-facing product scope.

Baseline results

Baseline checks: 31 | Success: 27 | Non-success: 4

After checks: 38 | Success: 37 | Non-success: 1

Before/After highlights

- New health endpoint added: before=n/a ms, after=14.75 ms
- New readiness endpoint added: before=n/a ms, after=20.94 ms
- Resume Builder improved timing and reliability: before=38679.6 ms, after=29499.58 ms
- Resume Optimizer remained stable with document outputs: before=40701.18 ms, after=31387.84 ms
- Portfolio generation completed under the test timeout: before=184525.75 ms, after=142558.07 ms
- Job application workflow completed instead of timing out: before=240046.67 ms, after=153347.43 ms

Architecture changes

- Added structured logging and request context utilities.
- Added centralized error handlers and safe API error payloads.
- Wrapped OpenAI chat completions with retry-aware proxy logic.
- Added health, readiness, database, AI, and storage endpoints.
- Added in-memory rate limiting and idempotency protection for expensive endpoints.
- Added admin monitoring endpoints for performance, latency, failures, and health summary.
- Added document validation for generated PDF and DOCX resume artifacts.
- Added regression tests covering health, admin protection, rate limiting, idempotency, error handling, route compatibility, and logging safety.

Known issues

- The unauthorized admin summary check appears as a non-success in raw metrics, but this is expected protection behavior.

- Long-running workflows such as portfolio and full job application are still the slowest paths and are candidates for future background-job offloading.

Acceptance result

- V2.0 status: Pass with known performance follow-up for the slowest orchestration flows.

Manual review folder: C:\Users\surya\AI-Jobs\performance_regression\after_v2_0\manual_review

Baseline folder: C:\Users\surya\AI-Jobs\performance_regression\baseline_v2_0

After folder: C:\Users\surya\AI-Jobs\performance_regression\after_v2_0